

Coded Beam Training

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Coded Beam Training for XL-MIMO in 6G

Problem Setup



- High Overhead
- Error Propagation

- Exhaustive Beam Sweeping

- Hierarchical Beam Sweeping

Key Idea

Beam Training ~ Channel Coding

Spatial Directions

0 0 1 1 1 0 1 0

Encoding

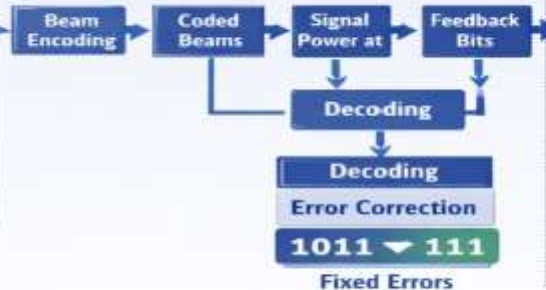
Hamming / Convolutional Code

1 1 0 1 1

Space-Time Beam Pattern



Coded Beam Training Process



Results / Advantages

- Exhaustive Beam Training: High Accuracy, High Cost



- Hierarchical Beam Training: Fast but Error-Prone



- Coded Beam Training: Efficient & Robust



- Higher Success Rate at Low SNR.

- Reduced Training Overhead

- Extended Coverage Area



Problems in Beam Training for XL-MIMO Systems



Exhaustive Beam Sweeping

Full Search Over All Beams



High Overhead



- Exhaustive Beam Sweeping
- Hierarchical Beam Sweeping

Hierarchical Beam Sweeping

Broad to Narrow Search



Error Propagation

Key Issues

- Extensive Training Time



- Risk of Incorrect Beam Selection



- Higher Success Rate of Low SNR

- Reduced Training Overhead

- Extended Coverage Area



Coded Beam Training Procedure (Flow)

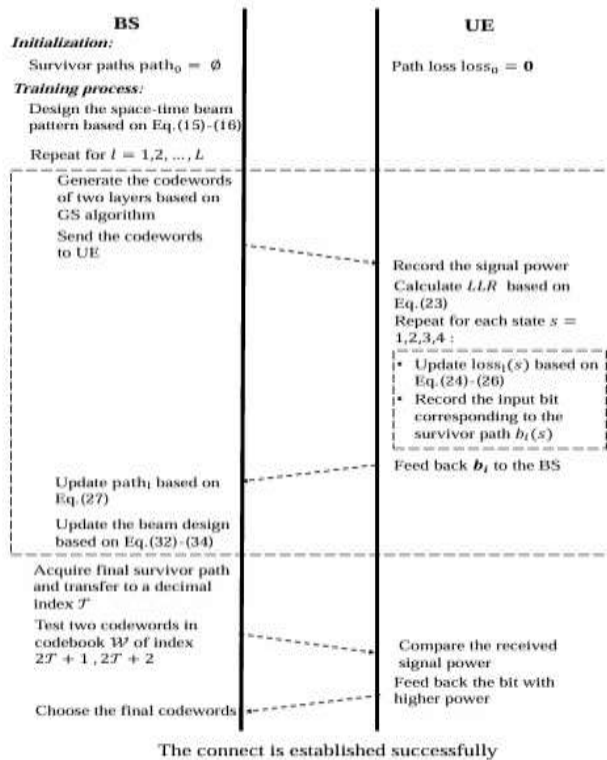


Fig. 7. Signaling diagram of the coded beam training procedure.

At Base Station (BS)

- Initialize: no prior path information
- Design space-time coded beam patterns
- Generate two-layer codewords (CS algorithm)
- Transmit coded beams over multiple stages

At User Equipment (UE)

- Measure received signal power
- Compute LLR (Log-Likelihood Ratio)
- Repeat for multiple stages ($s = 1, 2, 3, 4$)
- Update likelihood of beam indices
- Identify survivor path
- Send feedback to BS

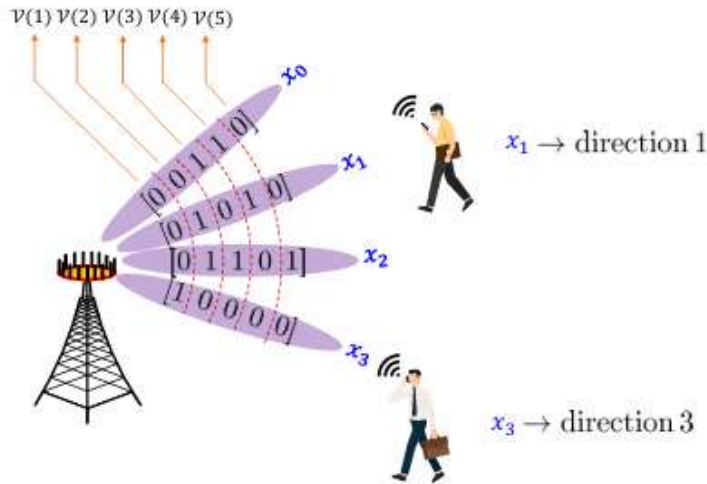
Refinement Phase

- BS updates beam design based on feedback
- Perform finer search (decimal index refinement)
- Test selected candidate beams

Final Step

- Select beam with maximum received power
- Establish communication link

Space-time beam pattern



Space-time beam pattern: directions are encoded into x and then beam pattern V is constructed based on it

Beam Decoding Procedure:

1. Codebook & Beam Assignment

- The base station (BS) uses the predefined codebook C
- Beamforming vectors are selected layer-by-layer: $C(l,1)$ and $C(l,2)$
- BS transmits pilot signals using these beam pairs

2. Signal Reception at UE

- User Equipment (UE) receives pilot signals
- UE measures received signal power for each beam:
 - $P(l,1)$ for first beam
 - $P(l,2)$ for second beam

3. Hard Decision (Bit Generation)

- UE compares received powers at each layer
- Decision rule:
 - If $P(l,1) > P(l,2) \rightarrow \text{bit} = 0$
 - Otherwise $\rightarrow \text{bit} = 1$
- A complete bit sequence $\hat{x}$ is formed after all layers

4. Feedback to BS

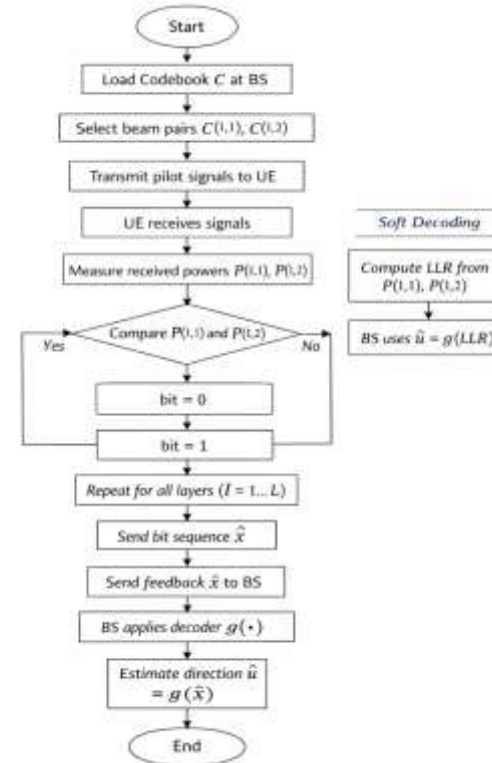
- UE feeds back the bit sequence $\hat{\mathbf{x}}$ to the BS
- This sequence represents the observed spatial direction information

5. Direction Estimation at BS (Hard Decoding)

- BS applies decoding function $\mathbf{g}(\cdot)$
- Estimated user direction is obtained
- This corresponds to angular direction estimation.

6. Soft Decoding Approach (Alternative)

- Instead of hard bits, UE computes **Log-Likelihood Ratio (LLR)**
- LLR is derived from received power values
- BS directly uses LLR for decoding
- Provides improved accuracy over hard decoding



- The proposed decoder uses a **Viterbi-based convolutional decoding algorithm** to efficiently recover the transmitted bit sequence by finding the most likely path through the trellis.
- It incorporates a **modified Log-Likelihood Ratio (LLR) computation based on a non-central χ^2 (chi-square) distribution**, improving detection accuracy under noisy channel conditions.

Map decoder

- MAP (Maximum A Posteriori) decoder selects the most probable transmitted beam based on received signal
- It evaluates all possible beam directions from the codebook
- Computes posterior probability for each beam given the received noisy signal
- Chooses the beam with the highest probability as the final output
- Improves beam selection accuracy in noisy wireless environments compared to simple max-power detection.

Example:

A transmitter sends one beam from a codebook: **B1, B2, B3**

Receiver gets a noisy signal, so it is not sure which beam was sent

MAP decoder checks which beam is most likely based on the received signal

Calculated probabilities: $P(B1 \mid \text{received}) = 0.2$,

$P(B2 \mid \text{received}) = 0.6$, $P(B3 \mid \text{received}) = 0.2$

Decision: MAP decoder selects **B2** (highest probability)

Final output: Best beam = **B2**

It does NOT just pick strongest signal. It picks the beam with the **highest probability of being correct**

ML decoder

- ML (Maximum Likelihood) decoder selects the beam that best explains the received signal
- It compares the received signal with all possible beams in the codebook
- It does NOT use prior probabilities, only the received data
- Chooses the beam with the highest likelihood value

Example:

- Codebook beams: B1, B2, B3
- Received noisy signal is observed

Likelihood values:

- $L(B1) = 0.3$
- $L(B2) = 0.7$
- $L(B3) = 0.5$

Decision:

- ML decoder selects **B2** (highest likelihood)

Final output:

- Best beam = **B2**

ML decoder = pick the beam that most likely produced the received signal

It ignores prior information and focuses only on signal matching

Chi-Square Distribution in Beam Training

- Wireless signals are generally modeled using a **Gaussian (normal) distribution**
- In beam training, we use **received power**, which is the squared magnitude of the signal: $|\text{signal}|^2$
- When a Gaussian random variable is squared, the resulting distribution becomes a **Chi-Square distribution**

$$y = hx + n$$

$$|y|^2$$

Received signal:

Received power:

Types: **Central χ^2 :** Only noise (no signal) **Non-Central χ^2 :** Signal + noise (real beam training case)

Beam training mainly uses **Non-Central χ^2 distribution** since both signal and noise are always present

Example

- Noise only $\rightarrow$ Central χ^2
- Beam + noise $\rightarrow$ Non-Central χ^2 (used for detecting best beam)

Issue in Existing Decoding

- Existing decoding algorithms assume **Gaussian channel models**
- Beam training observations are **not Gaussian (power-based)**
- This mismatch leads to performance issues

Problems Caused

- Poor decoding accuracy
- Incorrect beam selection
- Performance degradation at low SNR

Saleh–Valenzuela Channel Model

- Standard **clustered multipath channel model** for mmWave communications
- Represents propagation as **clusters of multipath components**
- Each cluster contains multiple rays with different delays and angles
- Widely used for modeling **sparse mmWave/6G channels**
- Captures realistic behavior of high-frequency wireless propagation

mmWave

e

Frequency Range: 30–300 GHz (5G / early 6G)

Channel Characteristics

- Sparse propagation (few dominant paths)
- Mostly **far-field** → planar wavefronts
- High path loss → needs directional beams
- Narrow beams with moderate sensitivity

Beam Training Implication

- Search space: **Angle only (1D)**
- Beamforming = **beam steering**
- Common methods:
 - Exhaustive search
 - Hierarchical search
 - Coded beam training

Terahertz

Frequency Range: 0.1–10 THz (6G target)

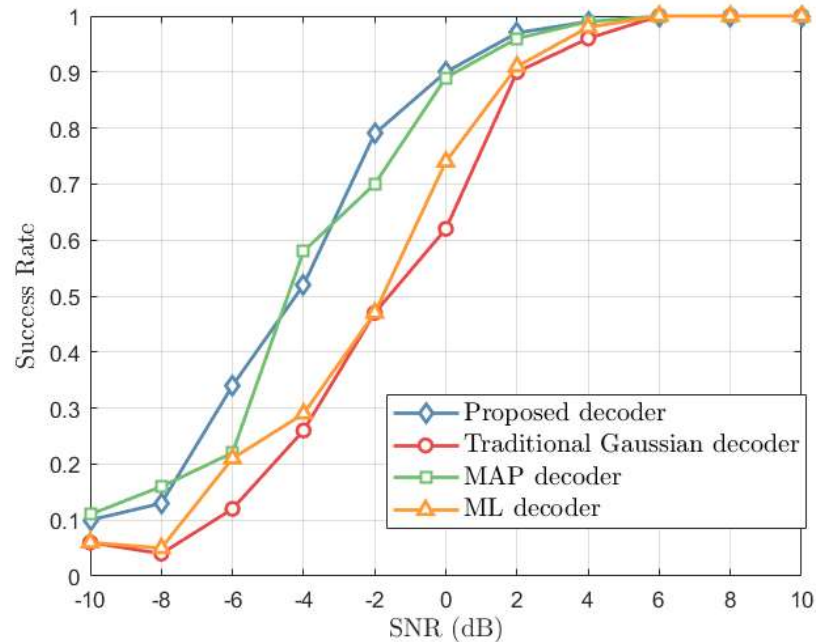
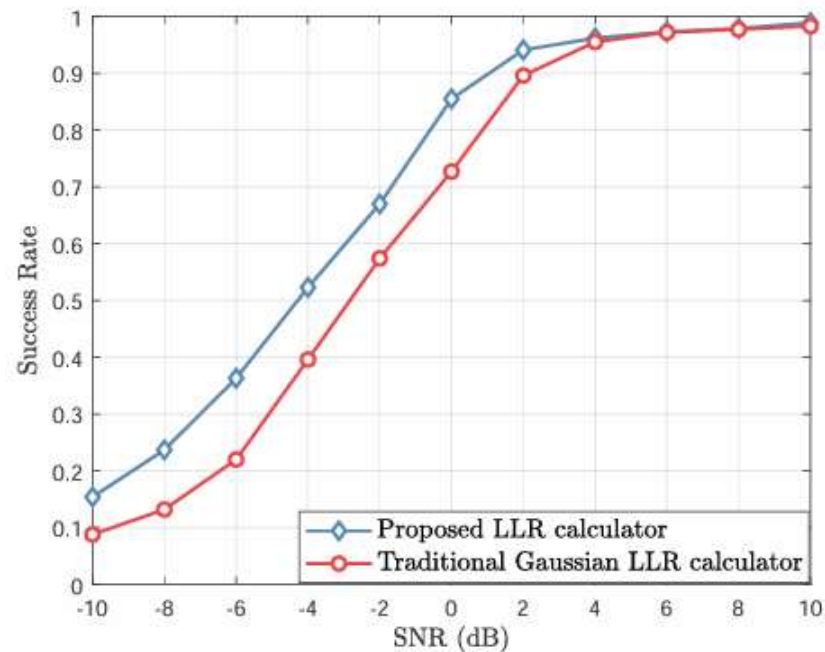
Channel Characteristics

- Ultra-sparse (often LOS dominant)
- **Near-field dominant** → spherical wavefronts
- Very high path loss + molecular absorption
- Ultra-narrow beams, highly sensitive

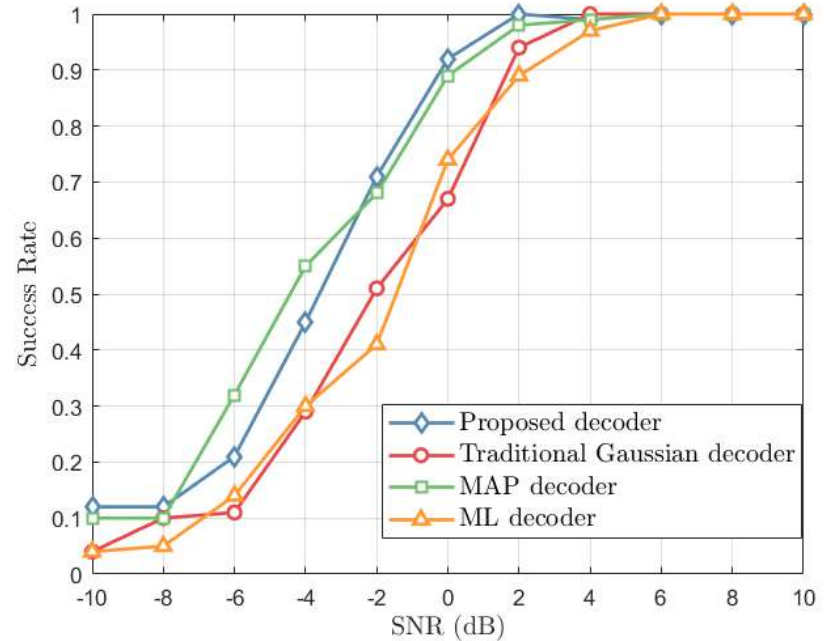
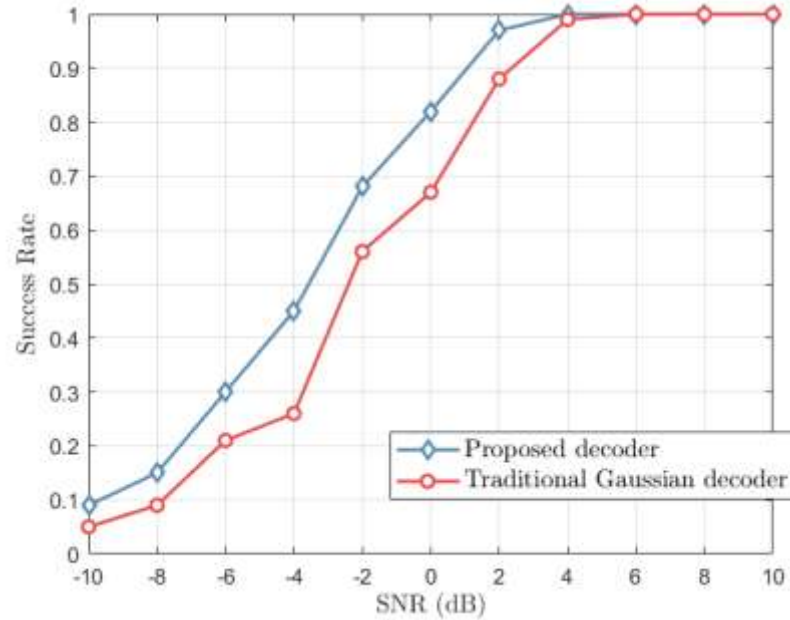
Beam Training Implication

- Search space: **Angle + Distance (2D)**
- Beamforming = **beam focusing**
- Requires:
 - Near-field (polar-domain) codebooks
 - Advanced methods (e.g., coded beam training)

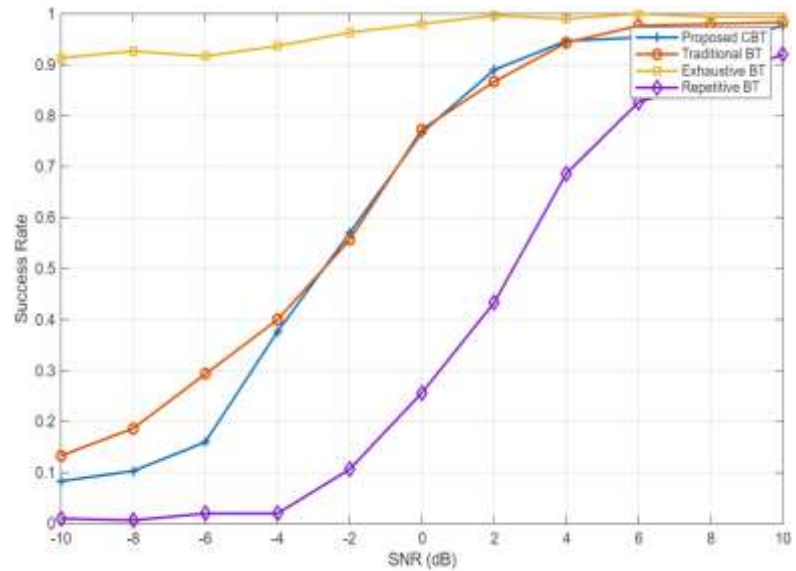
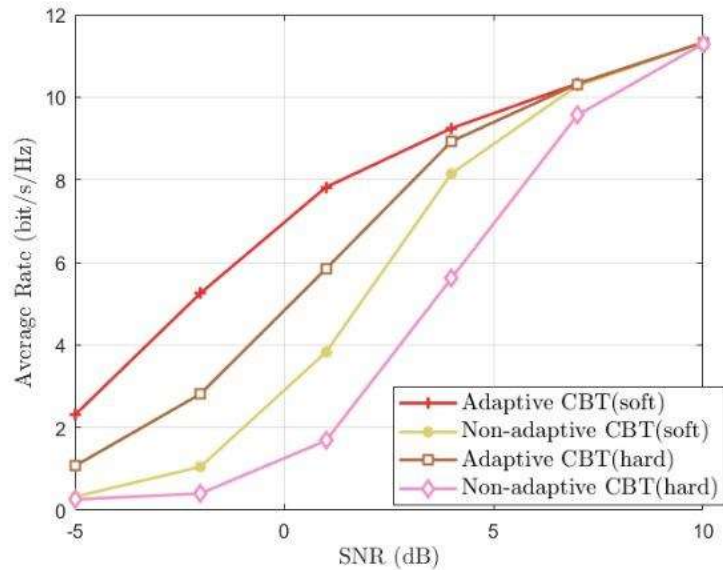
Results: 1) Decoder (mmWave)



2) Decoder (Terahertz)



3) Summrate (Terahertz)



THANK YOU!